MARITIME COMPLIANCE TOOLS

USER GUIDE

CII Calculator — Web Version

Carbon Intensity Indicator · Attained / Required CII

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

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1 Introduction

1.1 Purpose of this guide

This guide explains how to use the CII (AER) Calculator, a browser-based tool for calculating a vessel's Carbon Intensity Indicator (CII) under the Annual Efficiency Ratio (AER) method, in line with IMO resolution MEPC.354(78). It walks through every input field, the fuel consumption table, correction factors, results, rating bands, and the export/reporting functions, using screenshots taken directly from the tool.

The guide is written for ship operators, technical superintendents, and compliance officers who need to enter voyage and fuel data, run a CII calculation, interpret the resulting rating, and produce a shareable report.

1.2 What the tool does

- Calculates uncorrected and corrected CO_2 emissions from fuel consumption entries.
- Calculates Attained CII (AER) and compares it against the IMO Required CII for the vessel's type, size and reporting year.
- Assigns a CII rating band (A–E) and shows the boundary values used to determine it.
- Plots an interactive multi-year CII trend chart (2023–2030) against the rating boundaries.
- Supports optional correction factors (FCvoyage, TF, FCelectrical, FCboiler, FCothers, EEXI).
- Imports/exports fuel data via Excel, saves/reloads full cases as XML, and produces a print-ready PDF/Word report.

Before you start

- The tool runs entirely in your browser — no data is uploaded or sent anywhere.
- Have the vessel's Bunker Delivery Notes (BDNs), noon reports/logbook distance data, and IMO particulars ready.
- Use a current version of Chrome, Edge, or Firefox for the best experience.

2 Interface Overview

The calculator is organised into a fixed toolbar and four numbered sections: A — Vessel & Year Inputs, B — Fuel Consumption, C — Results, and D — Boundary Values. Each section can be collapsed using the arrow in its header.

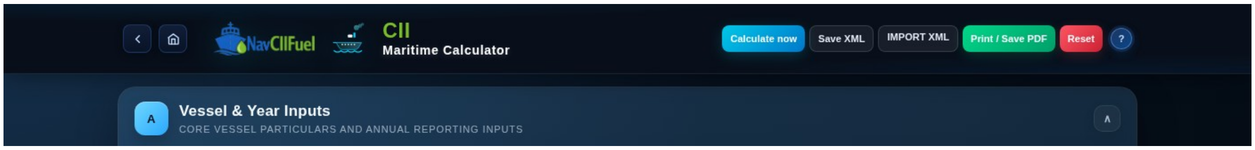

Figure 1 — Toolbar: Calculate now, Save XML, Import XML, and Print / Save PDF.

The toolbar buttons are available at all times and are described in the table below.

Button	Function
Calculate now	Runs the CII/AER calculation using the current inputs and refreshes Sections C and D.
Save XML	Exports all current inputs (vessel data, fuel rows, correction factors) as an .xml case file for later reuse.
Import XML	Loads a previously saved .xml case file and repopulates all fields automatically.
Print / Save PDF	Opens a formatted, print-ready compliance report and the browser print dialog (choose "Save as PDF" as the destination).
? (Page Help)	Opens the built-in on-screen help guide describing every field in Sections A–C.

3 Quick Start Workflow

Follow these six steps for a typical single-vessel, single-year CII calculation:

Step	What to do
1	Open the calculator HTML file in your browser and confirm the page loads with all four sections visible.
2	In Section A, enter the vessel name, IMO number, ship type, reporting year, capacity and annual distance sailed.
3	In Section B, add one row per fuel type consumed and enter the tonnes consumed for each, from BDNs and fuel records.
4	If applicable, switch on Correction Factors and configure only the factors that apply to the vessel.
5	Click Calculate now and review the CO_2, CII/AER, required CII, rating band and trend chart in Section C.
6	Use Save XML to archive the case and Print / Save PDF to generate the formal report for distribution.

4 Section A — Vessel & Year Inputs

Section A captures the vessel identity and the parameters that define the scope of the calculation: reporting year, capacity and distance sailed.

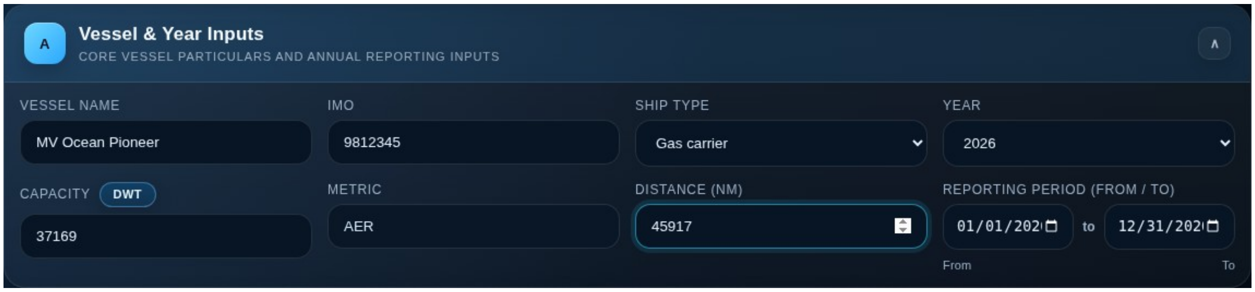

Figure 2 — Section A with sample vessel data entered.

Field	Description
Vessel Name	The name of the vessel being assessed. Used for labelling and the exported report.
IMO	The vessel's unique IMO identification number.
Ship Type	Selects the vessel category (e.g. Bulk Carrier, Tanker, Container Ship). Determines the capacity unit (DWT or GT) and the a/c coefficients used for the Required CII.
Year	The reporting year being assessed (2023–2030). Determines the Z reduction factor applied to the Required CII.
Capacity	The vessel's capacity in the unit shown by the badge next to the field — DWT for most ship types, GT for gas carriers and similar types set by the tool.
Distance (nm)	Total distance sailed during the reporting year, in nautical miles, from noon reports or the deck logbook.

Tip

- The Capacity unit badge (DWT/GT) updates automatically when you change Ship Type — always re-check the capacity value after switching ship types.
- Use the same reporting year consistently across Section A and any correction factor configuration.

5 Section B — Fuel Consumption

Section B records every fuel type consumed during the reporting year. CO_2 for each row is calculated as Consumption (t) × CF (tCO_2/tFuel).

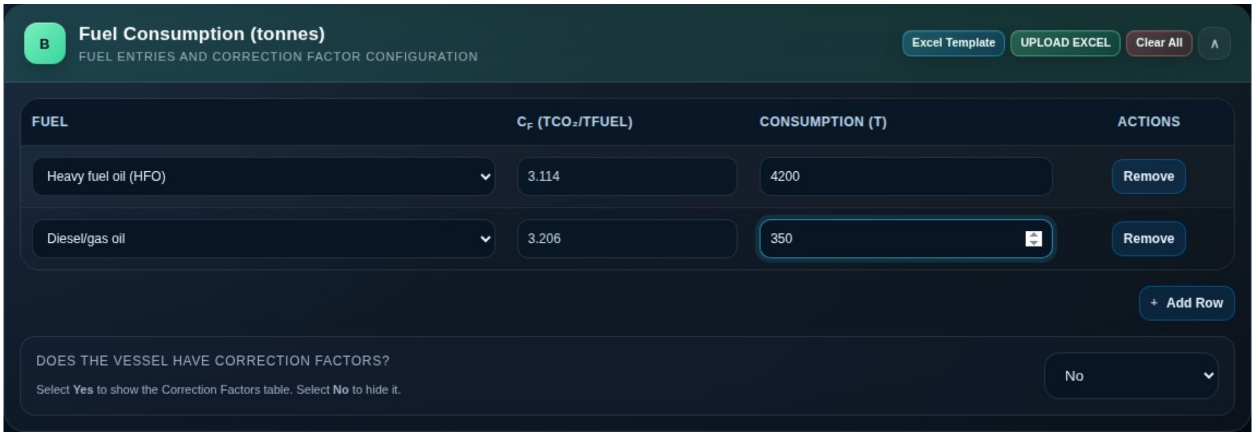

Figure 3 — Fuel Consumption table with two fuel rows entered (HFO and MGO).

Field / Control	Description
Fuel (column)	Fuel type selected from the drop-down (HFO, LFO, MDO, MGO, LNG, methanol, ethanol, LPG and more). Choose "Other" for a fuel not listed and enter its name manually.
CF (tCO_2/tFuel)	Carbon conversion factor for the selected fuel, per MEPC.354(78) Table 2. Auto-filled for standard fuels; enter manually for custom fuels.
Consumption (t)	Total mass of the fuel consumed in the reporting year, in metric tonnes, sourced from Bunker Delivery Notes and fuel records.
Add Row / Remove	Add Row appends a new blank fuel row; Remove (per row) deletes that row. There is no limit on the number of rows.
Excel Template	Downloads a blank .xlsx file with the correct column headers (Fuel, CF, Consumption) for offline data entry.
Upload Excel	Imports a completed .xlsx, .xls or .csv file and populates all fuel rows automatically, matching fuel names to the built-in list.
Clear All	Removes all fuel rows and replaces them with a single blank row — use to start the fuel section over.

5.1 Correction factors

Switch "Does the vessel have correction factors?" to Yes to reveal the correction factors table. Each factor can be marked applicable and configured individually via its Modify button; if left at No, the corrected values equal the uncorrected values.

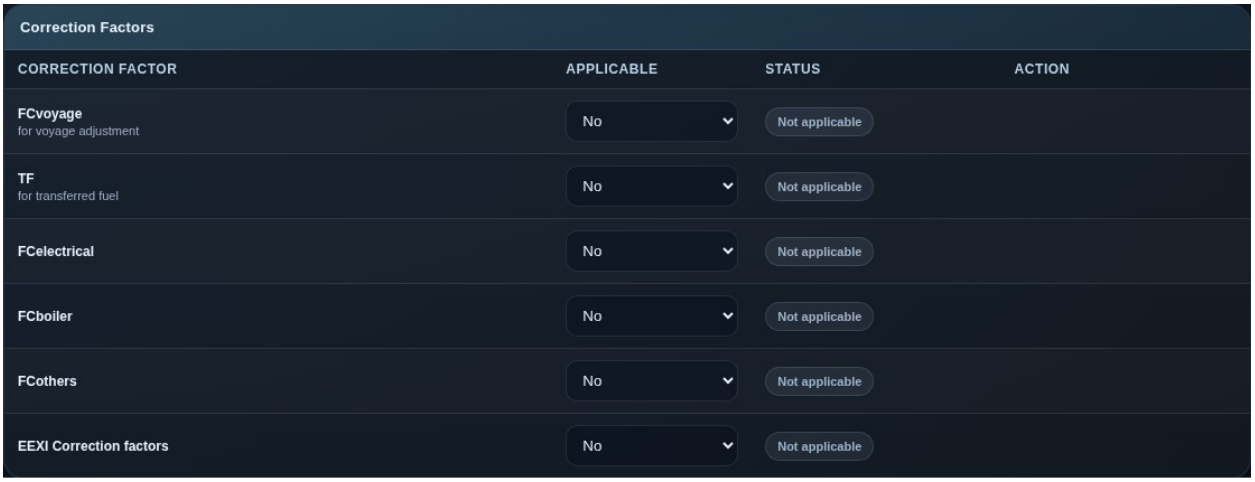

Figure 4 — Correction Factors table, shown after switching the toggle to Yes.

Correction factor	Applies to
FCvoyage	Voyage-specific distance/fuel adjustments (e.g. ice class or safety related deviations).
TF	Fuel transferred to or from other ships (e.g. STS operations), per the transferred-fuel formula in MEPC guidance.
FCelectrical	Fuel used to generate electrical power supplied to shore or other vessels.
FCboiler	Fuel consumed by boilers for approved technical reasons.
FCothers	Other approved technical adjustments not covered by the factors above.
EEXI	Correction factors sourced from the vessel's approved EEXI Technical File.

Important

- Only enable correction factors that are supported by the vessel's approved technical documentation.
- Re-run Calculate now after changing any correction factor so Sections C and D reflect the update.

6 Section C — Results

Section C appears after Calculate now is clicked. It requires no direct input — every value is derived automatically from Sections A and B.

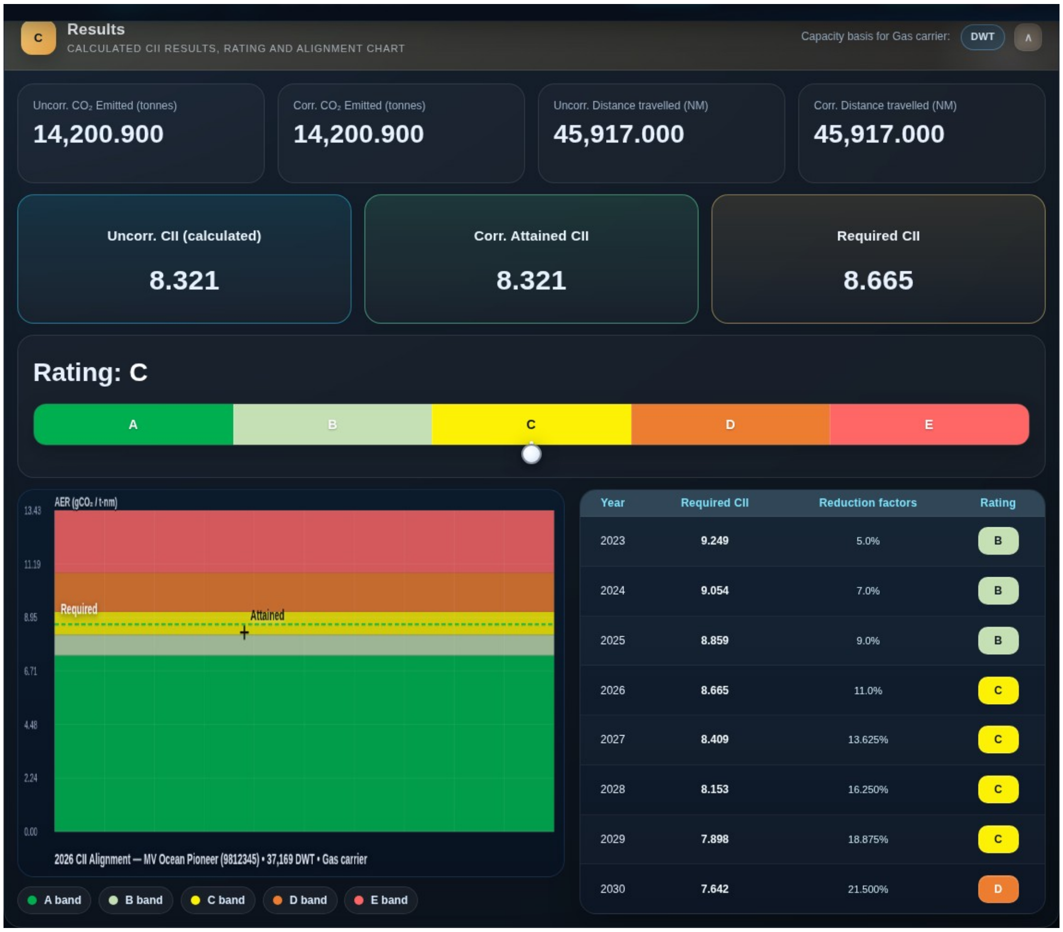

Figure 5 — Calculated KPIs, rating pill and CII trend chart for the sample vessel.

Result	Meaning
Uncorr. CO_2 Emitted (t)	Σ (Fuel × CF) for all rows — total CO_2 before any correction factors.
Corr. CO_2 Emitted (t)	CO_2 after applying all active correction factors; equals uncorrected CO_2 if none are set.
Uncorr. / Corr. Distance (NM)	Distance sailed before and after the FCvoyage correction, used as the denominator in the AER formula.
Uncorr. CII (Attained)	AER = CO_2 (g) ÷ (Capacity (t) × Distance (nm)), using uncorrected figures.
Corr. Attained CII	The official Attained CII, using corrected CO_2 and corrected distance — this is the figure used for rating comparison.
Required CII	($a \times Capacity^{-c}$) × (1 − Zyear/100), using the ship-type coefficients from MEPC.354(78) Table 1 and the year's reduction factor Z.
Rating (A–E)	Band assigned by comparing Attained CII to the Required CII boundaries d_1–d_4 (A = superior, E = inferior).

Result	Meaning
CII Trend Chart	Attained CII plotted against the A–E boundary lines for 2023–2030, with the selected year highlighted; hover for exact values.
Year Trend Table	Required CII, Attained CII and rating for every year 2023–2030 under the current inputs — useful for compliance-trajectory planning.

6.1 Boundary values

Section D lists the numeric CII boundaries (A ≤ … < B ≤ … < C ≤ … < D ≤ … < E) for the selected reporting year, letting you see exactly how close the vessel is to the next rating band.

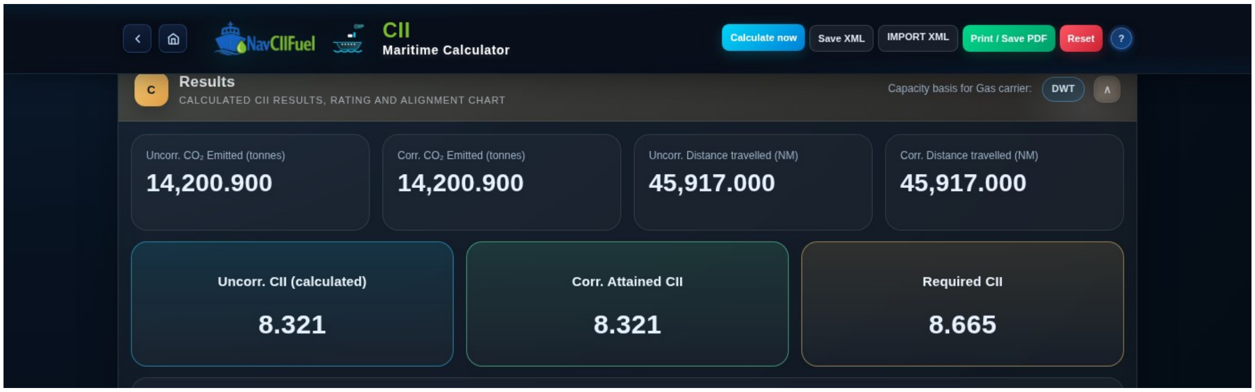

Figure 6 — Boundary Values for the selected reporting year.

7 Exporting and Record-Keeping

- Save XML after completing each case, to preserve every input value for future reuse or audit.
- Import XML to reload a saved case rather than re-entering data.
- Use Print / Save PDF once the results have been checked, choosing "Save as PDF" in the browser print dialog.
- Keep an editable Word copy of the report alongside the PDF for internal review and version tracking.
- Name saved files consistently, e.g. VesselName_IMO_ReportingYear_CII.xml / .pdf, for easy retrieval.

8 Best-Practice Checklist

- Use the same reporting year and reporting period consistently across all sections.
- Confirm capacity is entered in the correct unit (DWT or GT) for the selected ship type.
- Enter fuel consumption in tonnes and double-check CF values before calculating, especially for custom fuels.
- Apply correction factors only where supported by approved vessel technical documentation.
- After every calculation, review the rating band and boundary values before exporting the report.
- Store the XML input file together with the generated PDF/Word report for full traceability.

9 Troubleshooting

Symptom	Likely cause / fix
Results in Section C show "—"	Calculate now has not been run yet, or a required field in Section A is empty — fill all fields and click Calculate now.
Capacity unit shows the wrong badge (DWT/GT)	The Ship Type does not match the vessel — re-select the correct ship type; the badge updates automatically.
Corrected values equal uncorrected values	No correction factor is set to Yes — this is expected behaviour when no corrections apply.
Uploaded Excel file did not populate all rows	Fuel names in the file do not match the built-in list — unmatched fuels are set to "Other"; check and correct CF values manually.
Chart or trend table looks empty	Run Calculate now again after changing Year or Ship Type, as these affect the Required CII line.
Print / Save PDF opens a blank or partial report	Allow the browser a few seconds to render before printing, and ensure pop-ups are not blocked for this file.

A Appendix — Full Interface Reference

The overview below shows the complete calculator page with sample data entered and a calculation already run, for reference alongside the section-by-section descriptions in this guide.

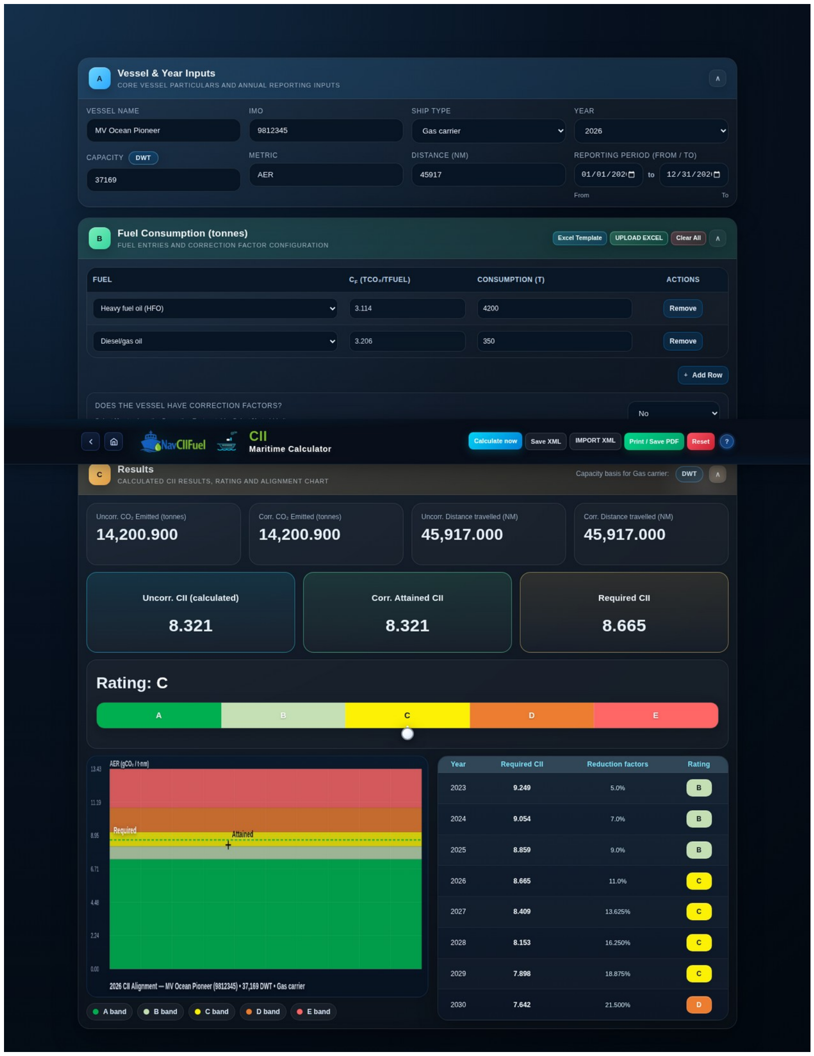

Figure 7 — Full calculator page after data entry and calculation.

End of user guide.